5.9.1 The composition and evolution of the Earth's atmosphere

5.9.1.1 The proportions of different gases in the atmosphere

Content	Key opportunities for skills development
For 200 million years, the proportions of different gases in the atmosphere have been much the same as they are today: • about four-fifths (approximately 80%) nitrogen • about one-fifth (approximately 20%) oxygen • small proportions of various other gases, including carbon dioxide, water vapour and noble gases.	MS 1c To use ratios, fractions and percentages.

5.9.1.2 The Earth's early atmosphere

Content	Key opportunities for skills development
Theories about what was in the Earth's early atmosphere and how the atmosphere was formed have changed and developed over time. Evidence for the early atmosphere is limited because of the time scale of 4.6 billion years. One theory suggests that during the first billion years of the Earth's existence there was intense volcanic activity that released gases that formed the early atmosphere and water vapour that condensed to form the oceans. At the start of this period the Earth's atmosphere may have been like the atmospheres of Mars and Venus today, consisting of mainly carbon dioxide with little or no oxygen gas. Volcanoes also produced nitrogen which gradually built up in the atmosphere and there may have been small proportions of methane and ammonia. When the oceans formed carbon dioxide dissolved in the water and carbonates were precipitated producing sediments, reducing the amount of carbon dioxide in the atmosphere. No knowledge of other theories is required. Students should be able to, given appropriate information, interpret evidence and evaluate different theories about the Earth's early atmosphere.	WS 1.1, 1.2, 1.3, 3.5, 3.6, 4.1

5.9.1.3 How oxygen increased

Content	Key opportunities for skills development
Algae and plants produced the oxygen that is now in the atmosphere by photosynthesis, which can be represented by the equation: $6CO_2 + 6H_2O \longrightarrow C_6H_{12}O_6 + 6O_2$ carbon dioxide + water $\xrightarrow{\text{light}}$ glucose + oxygen Algae first produced oxygen about 2.7 billion years ago and soon after this oxygen appeared in the atmosphere. Over the next billion years plants evolved and the percentage of oxygen gradually increased to a level that enabled animals to evolve.	WS 1.2 An opportunity to show that aquatic plants produce oxygen in daylight.

5.9.1.4 How carbon dioxide decreased

Content	Key opportunities for skills development
Algae and plants decreased the percentage of carbon dioxide in the atmosphere by photosynthesis. Carbon dioxide was also decreased by the formation of sedimentary rocks and fossil fuels that contain carbon.	
Students should be able to: • describe the main changes in the atmosphere over time and some of the likely causes of these changes • describe and explain the formation of deposits of limestone, coal, crude oil and natural gas.	WS 1.2, 4.1